

B2B SALES ORG DEVOPS PROCESS



October 28th, 2021

TABLE OF CONTENTS



B2B SALES ORG TEAM STRUCTURE



DEVOPS PROCESS OVERVIEW



DEPLOYMENT PIPELINE ARCHITECTURE



COPADO USERSTORY DEPLOYMENT READINESS



QUALITY GATES



SPRINT EXECUTION TIMELINE



SIT & UAT DEPLOYMENT ISSUES



DEVOPS PROCESS IMPROVEMENTS

B2B SALES ORG TEAM STRUCTURE

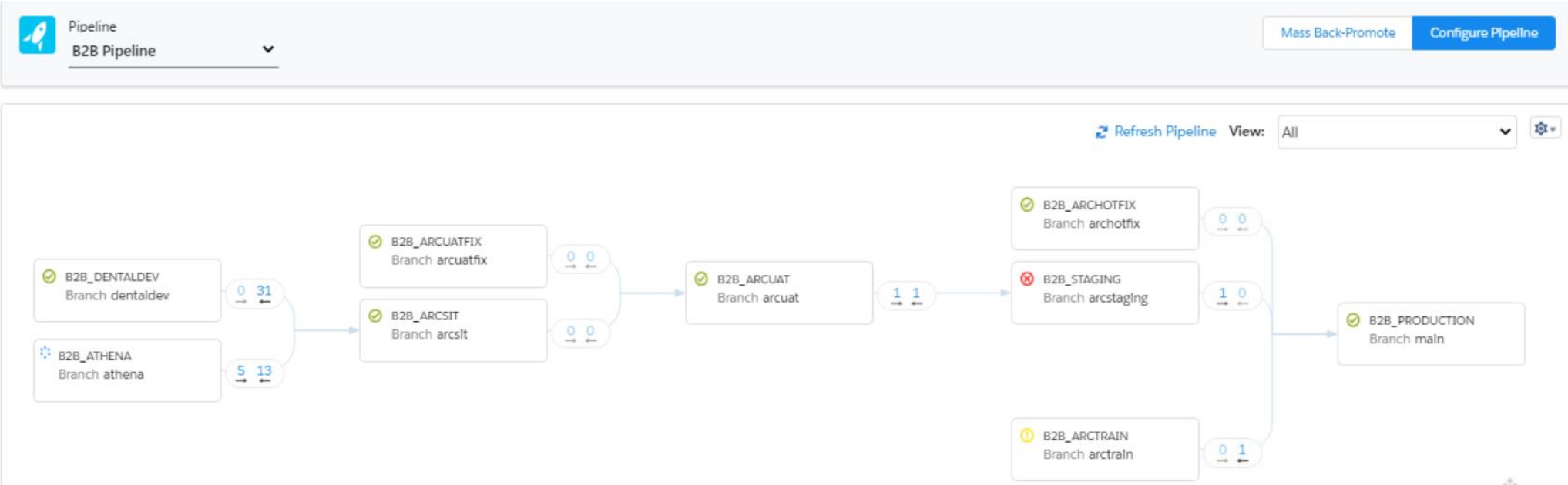
- **Teams**
 - **Group CRM**
 - Specialty Quoting Team 1
 - Opportunity Rangers
 - AI Integrators
 - The Jetsons
 - Above PAR
 - RAApers
 - **Dental**
 - Dental Network

DEVOPS PROCESS OVERVIEW

- **DevOps Tools**
 - **Azure Board**
 - **Azure Repo**
 - **Azure Pipeline**
 - **Copado**
- **Tools Integration:**
 - ✓ **Azure board** – Azure board for Sprint Planning, User Story and Bug Fix tracking for B2B Sales Org
 - ✓ **Azure Repo** – Azure repo used for source code management
 - ✓ **Azure Pipeline** – For Mulesoft deployment
 - ✓ **Copado Pipeline** – B2B end-to-end pipeline management
 - ✓ **Azure/Copado Sync** – To sync User Stories between Azure and Copado
 - ✓ Scheduled batch job at 12AM EST
 - ✓ Manual Sync as needed

DEPLOYMENT PIPELINE ARCHITECTURE - COPADO

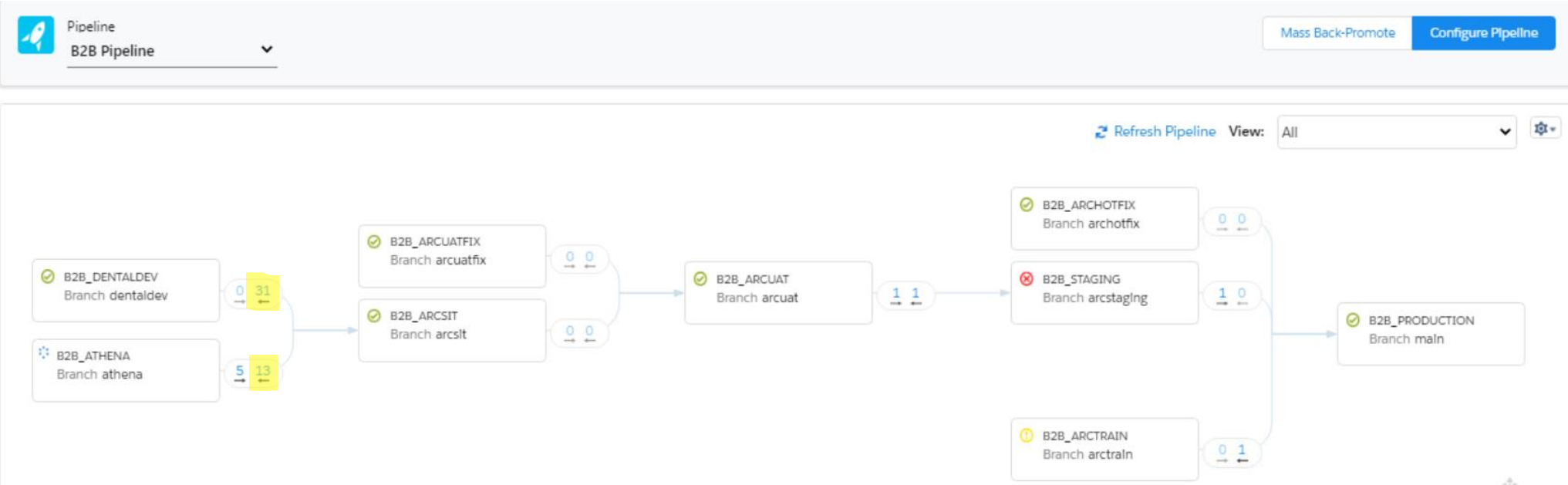
Copado User Story Progression Path



Forward Promotion	Dev-to-SIT	SIT-to-UAT	UAT-to-Staging	Staging-to-Production
Responsibility	Developer	DevOps	DevOps	DevOps

DEPLOYMENT PIPELINE ARCHITECTURE - COPADO

Copado User Story Back-Promotion



Back-Promotion	SIT-to-Dev	UAT- UATFix	Prod-Hotfix/Arctrain
Responsibility	Developer	DevOps	DevOps

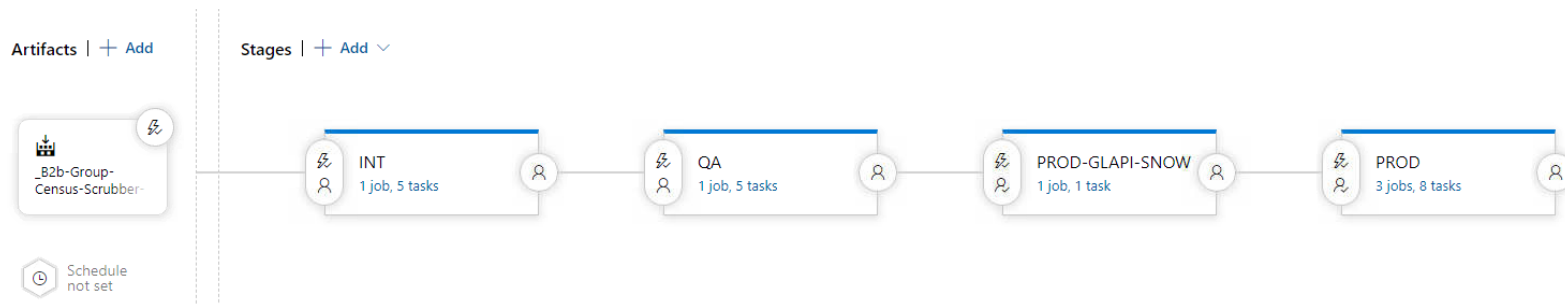
DEPLOYMENT PIPELINE ARCHITECTURE - AZURE

- **Mulesoft Build Pipelines**

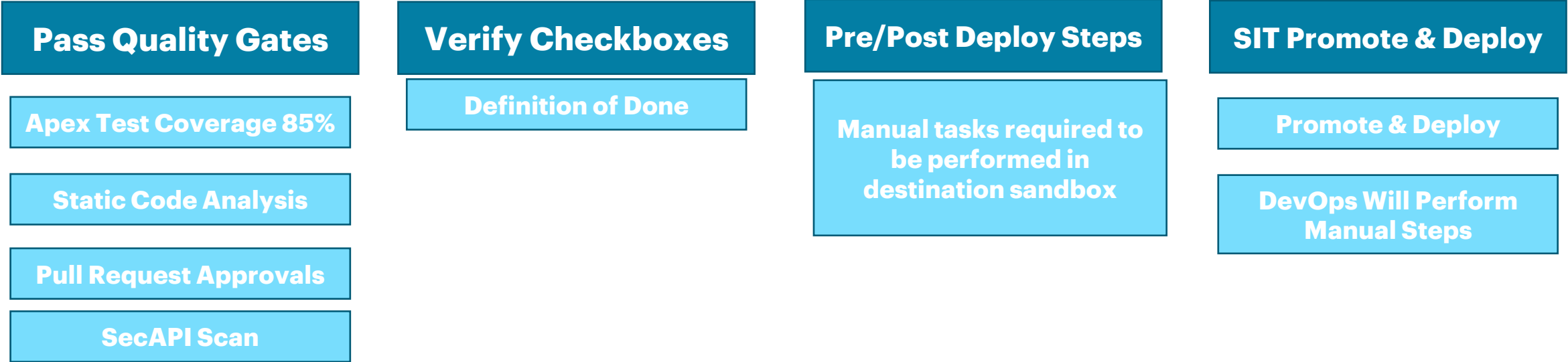
- ✓ **B2B-Group-Census-Scrubber-sfdc-Build-Pipeline-develop:**
Builds artefact from develop branch and deploy to Dev environment
- ✓ **B2B-Group-Census-Scrubber-sfdc-Build-Pipeline-master:**
Builds artefact from master branch and trigger release pipeline

- **Mulesoft Release Pipeline**

- ✓ **B2B-Group-Census-Scrubber-sfdc-Release-Pipeline:**
Deploys build artefact to SIT, UAT and Production



COPADO USER STORY DEPLOYMENT READINESS



QUALITY GATES

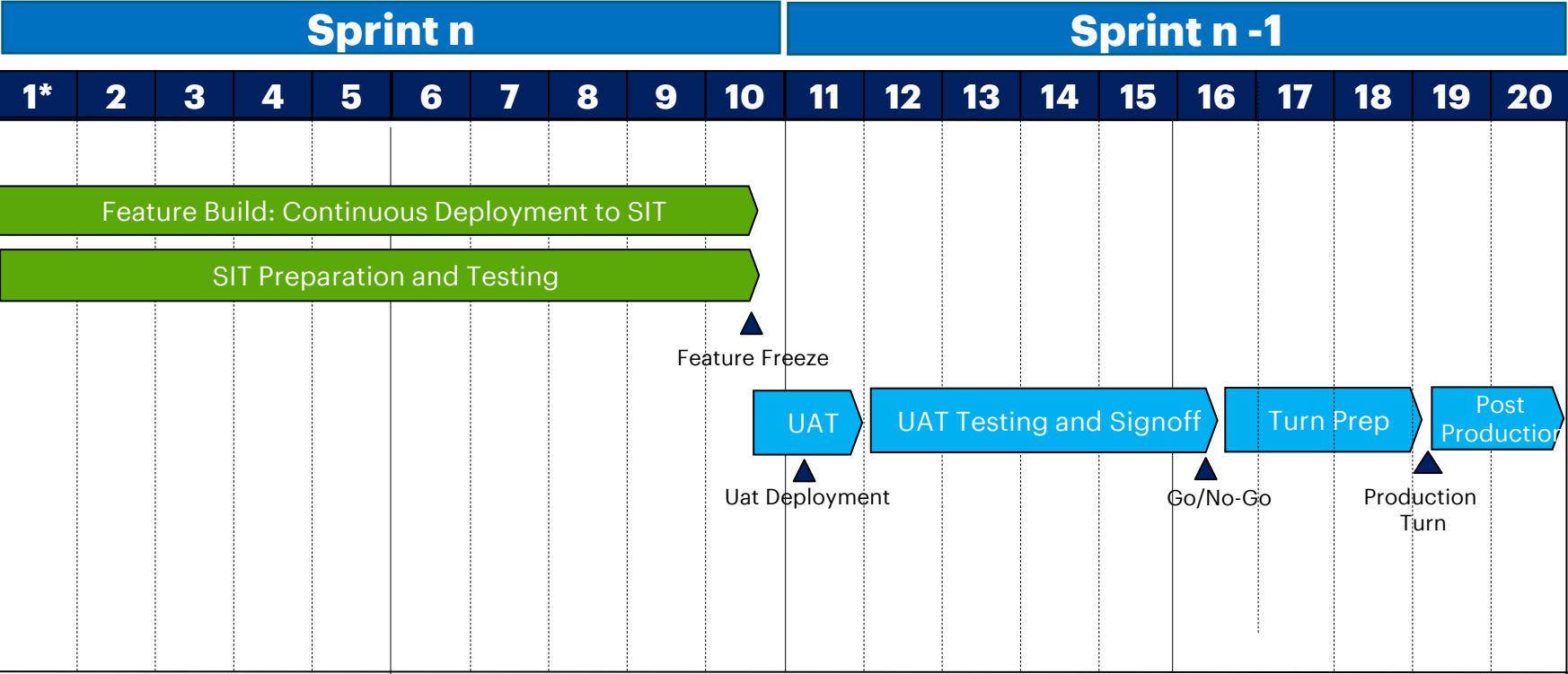
✓ Copado User Story Level

- ❖ Apex test coverage of 85%
- ❖ SCA – Priority 1 violations will block deployments
- ❖ Pull request Approvals
 - ❖ 1 Level of Approval if story has no Security components (Approval from Dev Lead)
 - ❖ 2 Levels of Approval if story has Security components (Approval from Dev Lead & Security Team)
- ❖ SecAPI scan

✓ Copado Pipeline Level

- ❖ Dev-to-SIT
 - ❖ Stories with Apex Components to go through Run All Tests in SIT
 - ❖ Stories without Apex components run no tests
- ❖ DevOps runs a weekly scan of all test classes in SIT and UAT using Copado
- ❖ Manual SecAPI scan at branch level for Main and UAT (Performed by Security Team)
- ❖ GLAPI process will be running a scan at Release level, replacing the manual scan requirement mentioned above

SPRINT EXECUTION TIMELINES



**indicates business days*

Cut off Time Guide

- ✓ Stories must be in 'SIT Done' Azure Board column by **Friday, 2 PM EST on the last day of the sprint** to be picked for UAT deployment.
- ✓ Stories must be deployed to UAT between **Friday, 2 PM EST** and following **Monday 8PM EST**.
- ✓ UAT Testing starts by following **Tuesday 2 AM EST**.
- ✓ Stories must be in '**Pending release**' Azure board column by **Monday 5PM of the release week** to be included in the release package.

SIT & UAT Deployment Issues

What Causes Delays

✓ **Deployment Issues**

- ❖ Most deployment errors are caused by incorrect commits or merge conflict resolutions. Deployment issues can be reduced by strictly following below measures.
 - ❖ Dev Leads to review and approve pull requests
 - ❖ Developer to verify feature and promotion branch
 - ❖ Developer to perform smoke test immediately after deployment

✓ **Copado Support Issues**

- ❖ When DevOps/Security/Developers are unfamiliar with an error and unable to resolve, DevOps reach out to E3 for Copado Support
- ❖ There is a wait time for support to come through and resolution is usually a time taking process

DevOps Process Improvements

How DevOps can automate our workflow to improve efficiency

✓ Back-Promotions

- ❖ Automated/scheduled back promotion jobs to regularly back-promote stories to ensure development box integrity. This process significantly improve efficiency of the process and reduce merge conflicts.
- ❖ Automated process is still a work-in-progress due to a Copado bug. Meanwhile DevOps regularly perform audits to ensure manual back promotions are completed as part of the sprint.

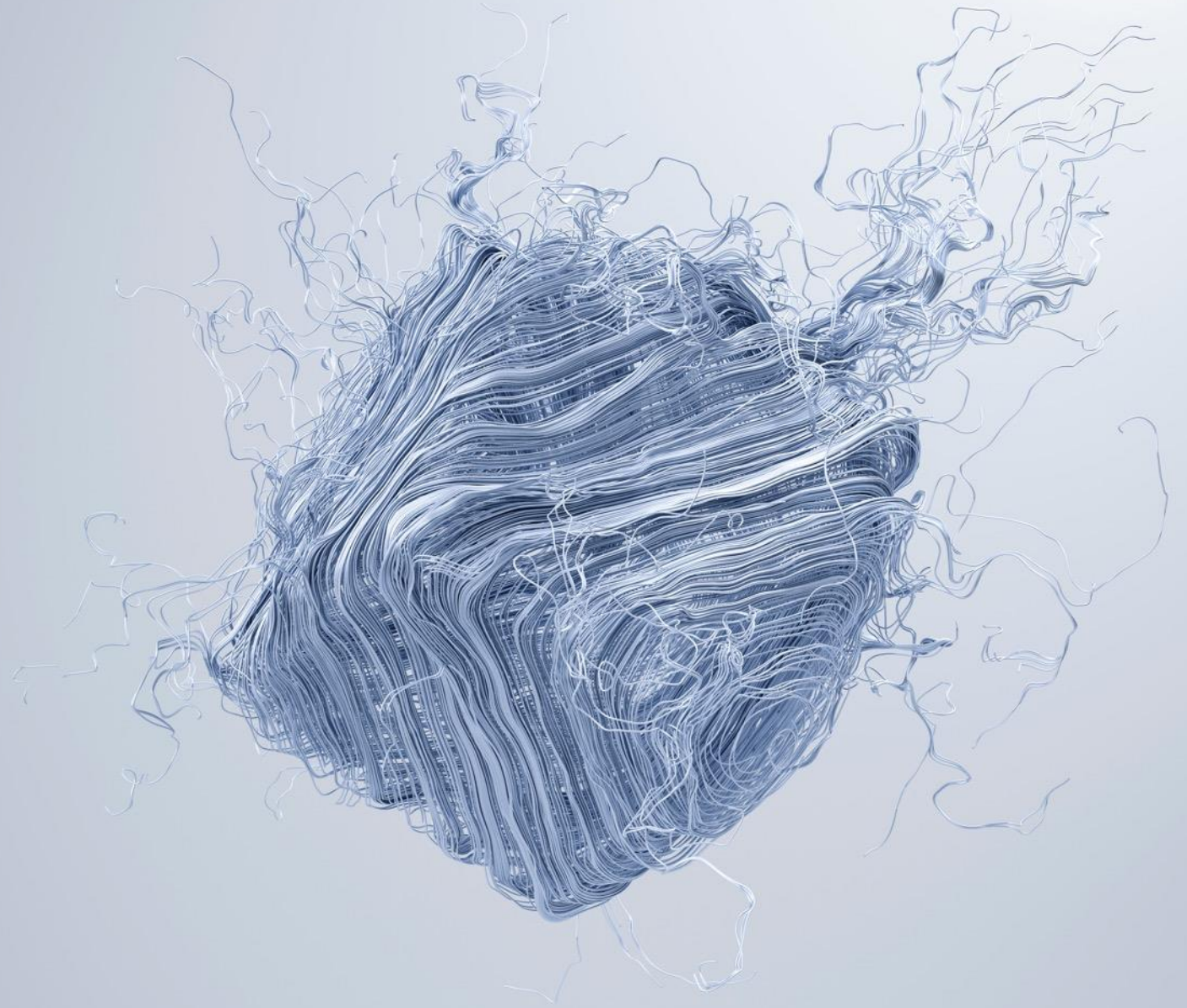
✓ Intake Process

- ❖ Currently DevOps support need a lot of back-and-forth over email and teams without the ability to properly track task status
- ❖ DevOps intake process will help streamline requests coming to DevOps queue and significantly improve effectiveness of support.
- ❖ Intake process will help DevOps metrics report and tracking



Q & A

APPENDIX



What To Master in Source Control

Many post deployment steps can be included in source control because the files these changes reside in are available for import through the metadata API, the interface which allows extraction from SFDC environments to Copado. In reviewing the changes executed in production these past few months, there are opportunities to automate.

Mastered in Source Control

- Applications, App Menus, **Approval processes**, aura, auth providers auto response rules, **assignment rules**
- **Classes**, components, communities, connected apps, custom application components, custom meta data, custom labels, custom permissions, Data category groups, documents
- **Escalation Rules**, email, flexipages, **Flows**, groups, home page components & layouts, **Layouts**, Letterhead
- **Objects – all custom and standard**, object translations
- **Pages, permission sets**, Post Templates, **profiles**
- **Queues**, quick actions, remote site settings, roles
- SAML SSO Configs, sControls, settings, sitedotcoms, sites, static resources, **Triggers**, tabs, test suites, **workflows**, weblinks
- **Vlocity**: AttributeCategory, CalculationProcedure, **DataRaptor**, EntityFilter, IntegrationProcedure, ItemImplementation, **OmniScript**, OrchestrationItemDefinition, OrchestrationPlanDefinition, Pricebook2, Product2, StoryObjectConfiguration, System, VlocityAction, VlocityCard, VlocitySearchWidgetSetup, VlocityUILayout, VlocityUITemplate

Mastered in Production

- Reports
- Dashboards
- Certificates
- Static Resources
- Environment-specific variables (e.g., custom settings, URLs, etc.)
- List Views
- Managed Packages
- Assigning users to Public Groups
- Vlocity Calculation Matrix
- Vlocity Calculation Matrix Version
- [Unsupported Metadata Types](#)

MERGE CONFLICTS BEST PRACTICES

Merge Conflicts

When there is a merge conflict Copado provides 2 options:

1) Copado Conflict Auto-Resolution

- ✓ Copado uses Semantic merge for XML based files. For all other files, A wins over B
- ✓ Git repository keeps documentation of auto-resolved conflicts and can be reverted easily.

2) **Online Conflict Resolution**

- ✓ Copado allows certain metadata types to be excluded from auto resolution.
- ✓ This can be set up either at pipeline or Promotion level
- ✓ This allows developers to review and manually resolve conflict

Smart Conflict Resolution feature

Copado will learn how each conflict have been resolved without prompting you again. The conflict will be marked as AUTO RESOLVED

Reference Links

- ✓ **E3 site for Copado Documentation and DevOps process:** [go/learncopado](#)
- ✓ **Static Code Analysis (SCA)** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/pmdpr/>
- ✓ **Pull Requests** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/pr/>
- ✓ **SecAPI**: <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/secapi/>
- ✓ **FVM** : <https://devopsdocs.apps.sdc-prod-cf.humana.com/products/copado/enhancements/fvm/>